Introduction to Quantum Computing

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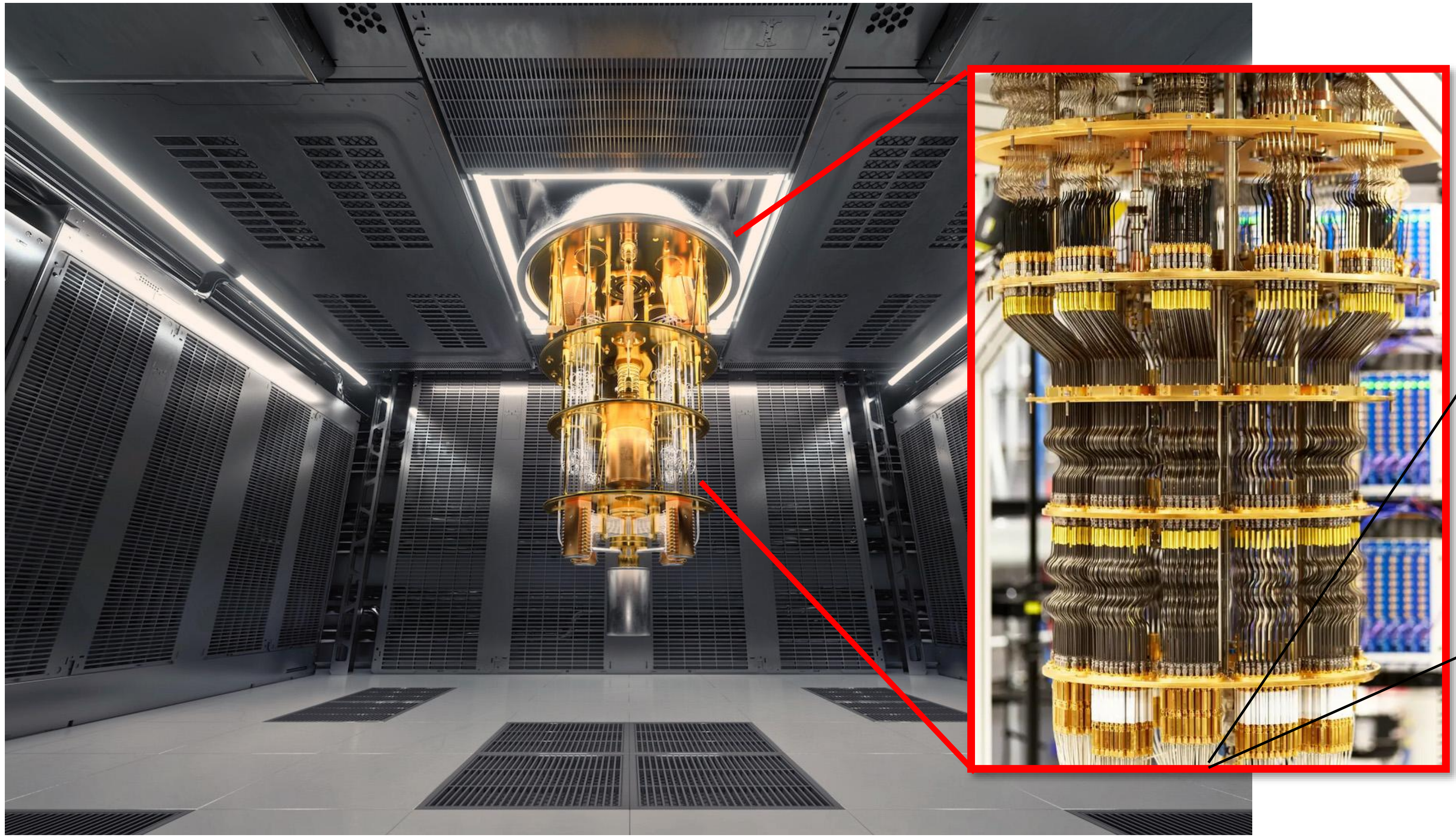
Introduction to Quantum Computing: Course Structure

- The Quantum World (L1)
- Mathematical Foundations and Recap of Linear Algebra (L2)
- Quantum Gates and Introduction to IBM Qiskit (L3)
- Entanglement, Teleportation, and Super Dense Coding (L4)
- Introduction to Quantum Algorithms – Deutsch- Jozsa Algorithm (L5)
- Simon’s Algorithm (L6)
- Quantum Fourier Transform and Period Finding Algorithm (L7)
- Shor’s and Grover Algorithms (L8)
- External Lecture – IBM Scientist on Noisy intermediate-scale quantum computing (L9)
- External Lecture – IIT Bombay Faculty on Quantum Approximate Optimization Algorithm (L10)

Introduction to Quantum Computing: Grade Structure

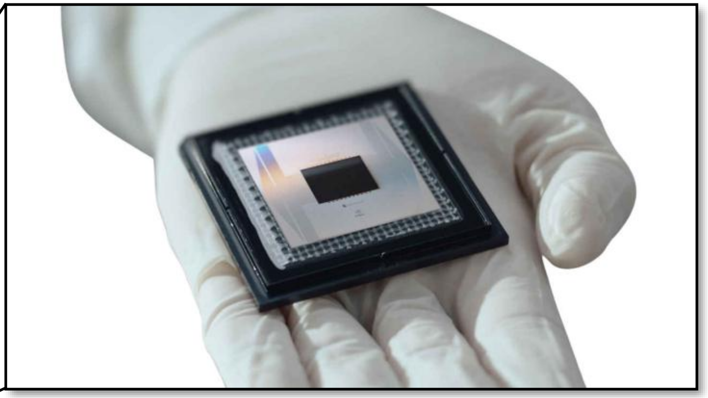
- Attendance + Participation in Class Discussions: 10%
- 2 Assignments – Qiskit programming of algorithms discussed in class: 30%
- 3 Quizzes – Concepts and Small Problems: 30%
- Project – Full Algorithm development from scratch: 30%

Introduction to Quantum Computing



This is not a computer in the true sense. It's a cryogenic physics experiment, pretending to do computation.

Quantum computers operate at ~10 millikelvin. 100× colder than deep space.



Quantum bits or Qubits Hate the World

The Gold Isn't for Luxury

The real computer is invisible

Why does a Quantum Computer look so strange?

Introduction to Quantum Computing

What does a “quantum” mean?

The word “quantum” in a quantum computer originates from quantum mechanics, a basic theory of physics.

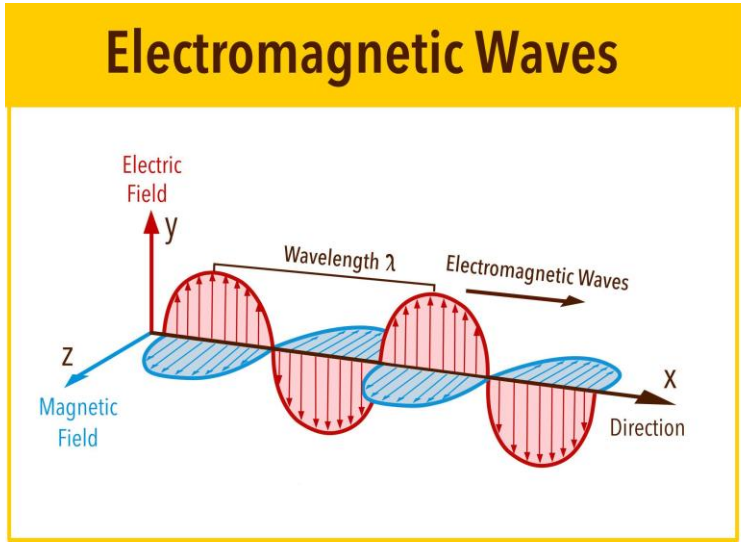
In brief, on the scale of electrons, atoms, & molecules, matter behaves in a quantum manner, obeying the laws of quantum mechanics, which is different from our everyday perception of the classical world

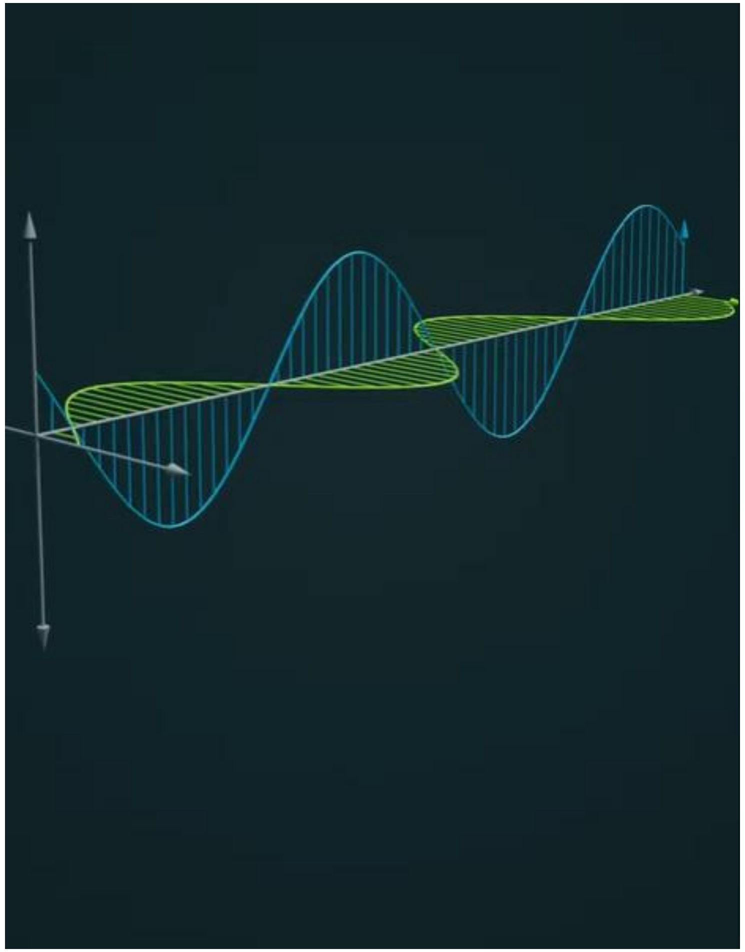
What is a Quantum Computer?

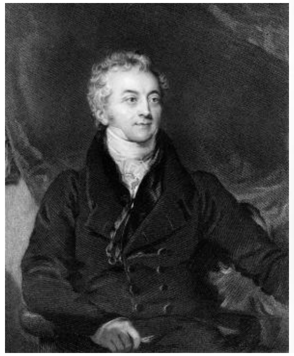
A quantum computer is therefore a machine that can perform calculations based on the laws of quantum mechanics, which is the behaviour of the particles at the sub-atomic level.

A Few Key Takeaways from the Quantum World

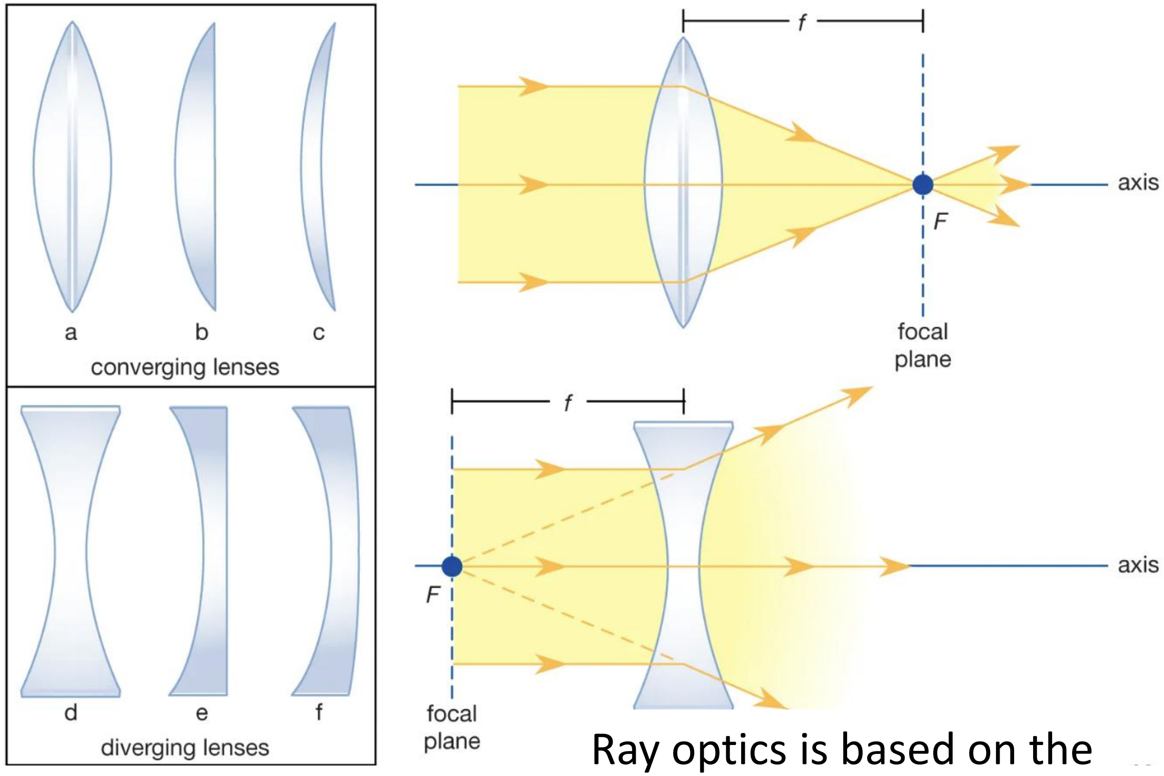
What is light?



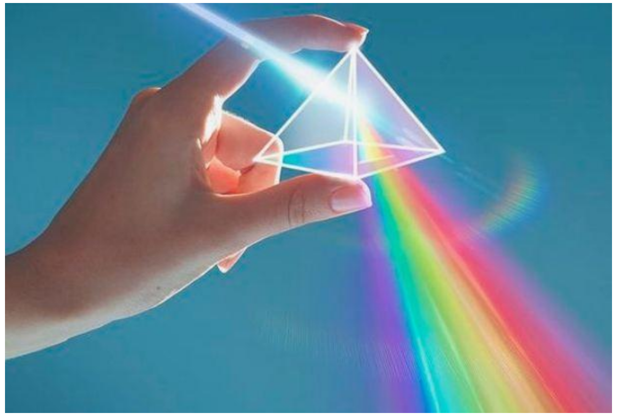


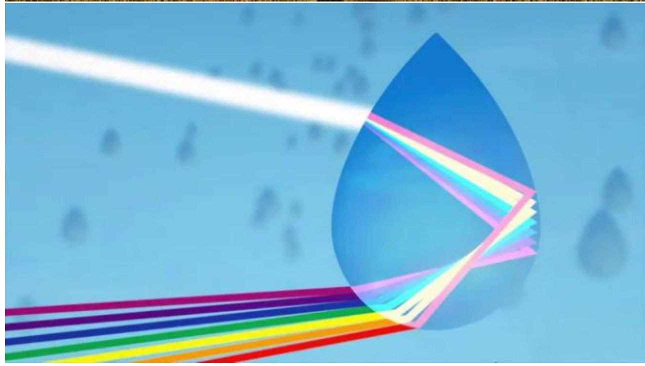
			
Christiaan Huygens (Deutch)	Thomas Young (British)	James Clerk Maxwell (Scottish)	Augustin-Jean Fresnel (French)

A Few Key Takeaways from the Quantum World - Waves



Ray optics is based on the wave nature of light





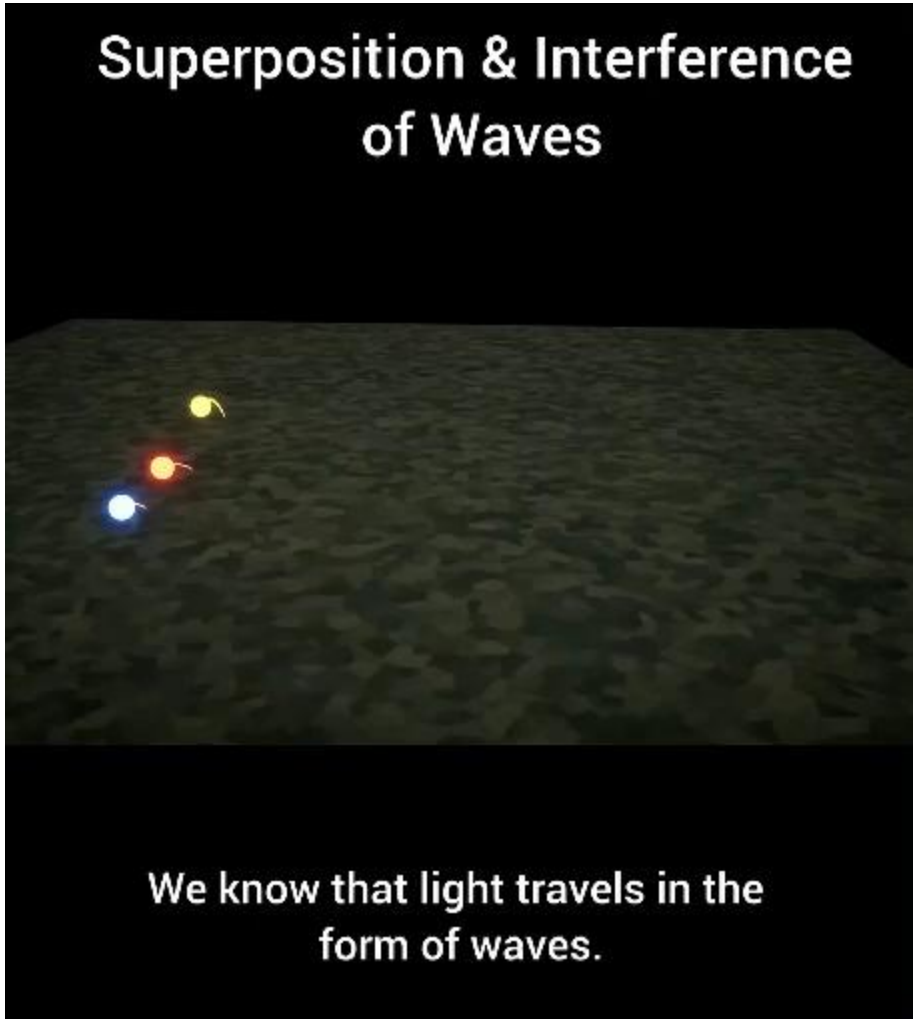
Dispersion is due to the wave nature of light

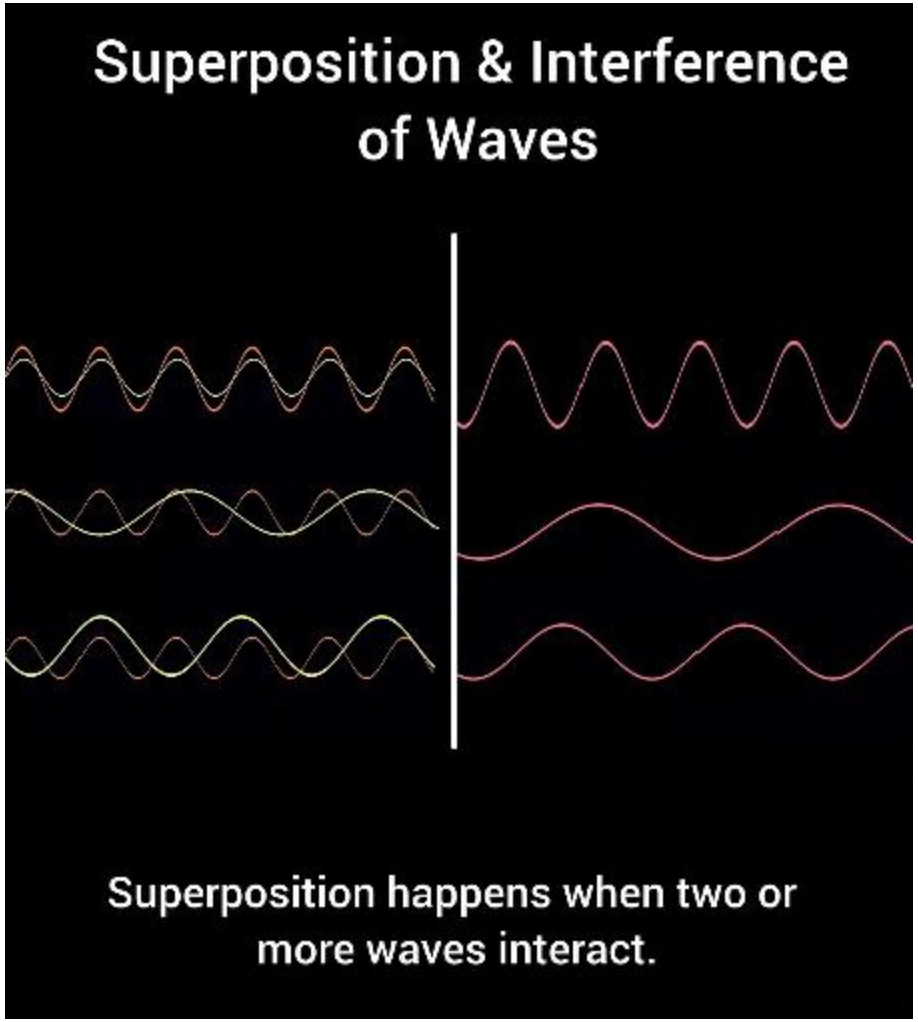
A branch of science called the Wave-Optics was developed based on the fact that light is an EM wave. This is something we all studied in our school science books

Diffraction – Bending of light around the edges of a sharp objects

A Few Key Takeaways from the Quantum World - Waves

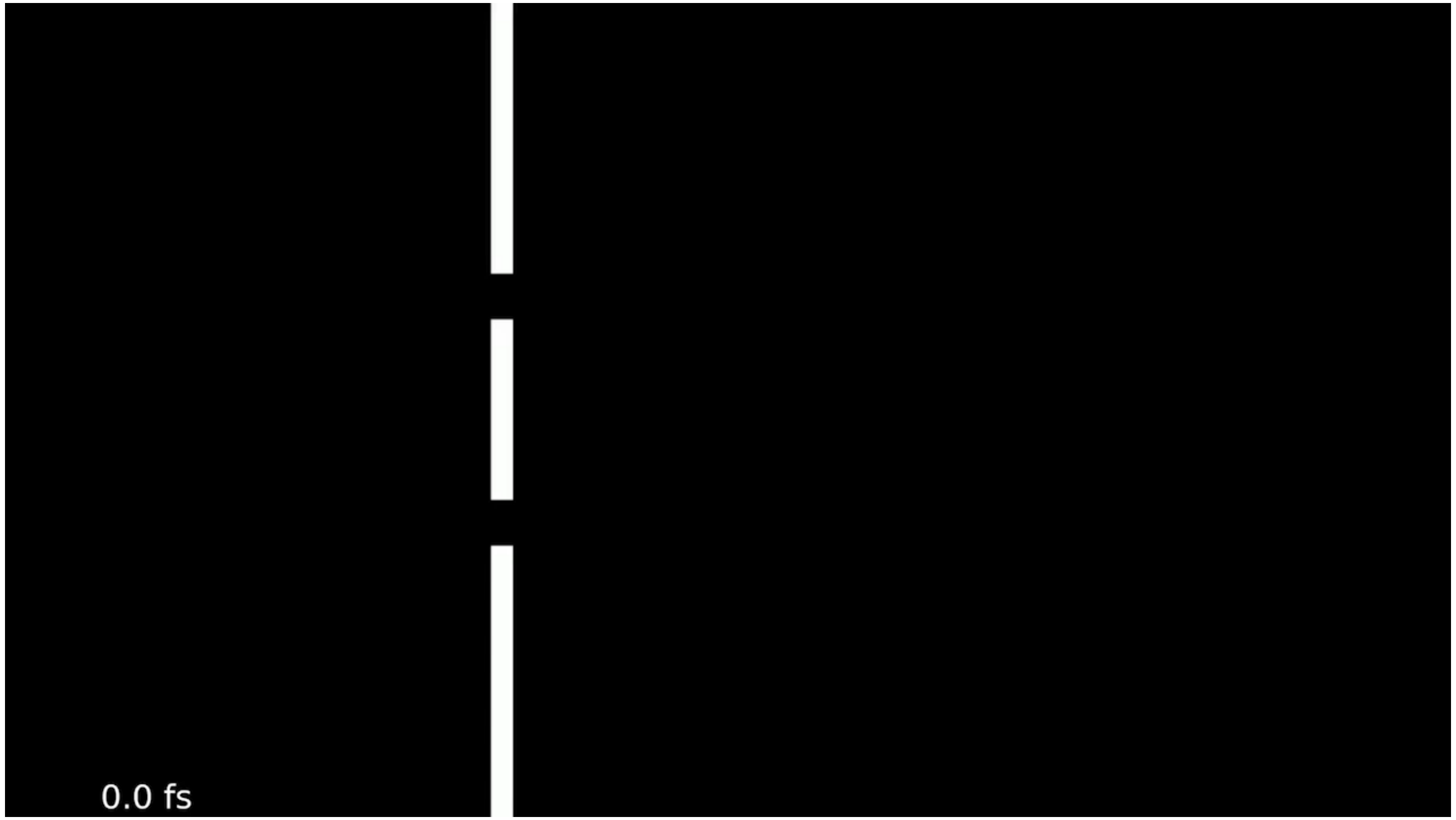
A Few Key Takeaways from the Quantum World - Waves





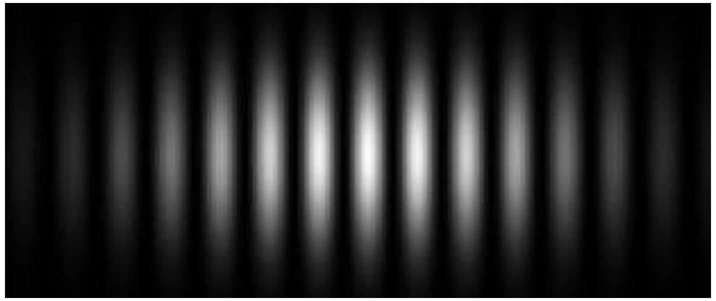
Supposition of waves and wave interference

A Few Key Takeaways from the Quantum World – The Double Slit Experiment



Young's double slit experiment – simulations results

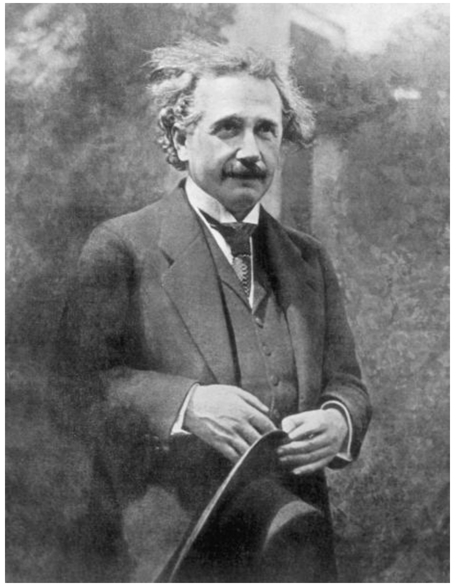
Experimental result



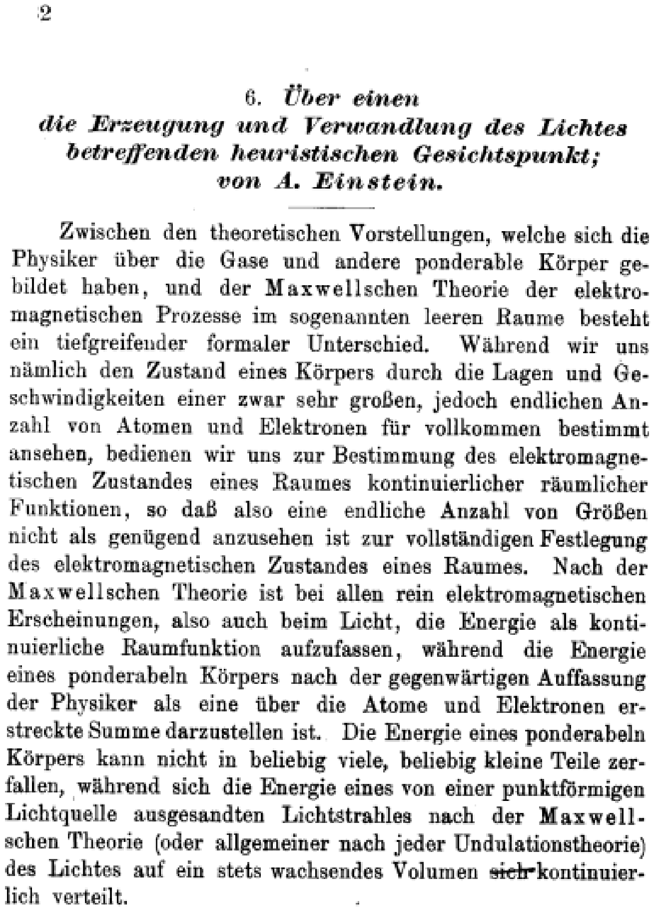
- When waves pass through two slits, they overlap and interfere with each other.
- On a screen behind the slits, we see a pattern of bright and dark bands.
- These bands are called interference fringes.
- Bright bands appear where waves add together (constructive interference).
- Dark bands appear where waves cancel each other (destructive interference).
- The double slit experiment was considered the "gold standard" test: If you see an interference pattern, then what passed through the slits must be a wave

A Few Key Takeaways from the Quantum World – The Particle Nature of Light

By 1900, scientists were confident:
Light is a wave. The debate seemed over.



Albert Einstein



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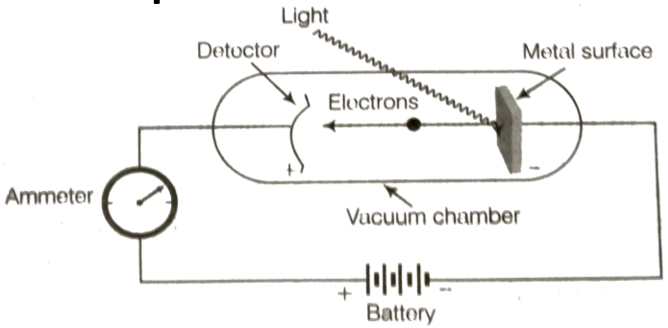
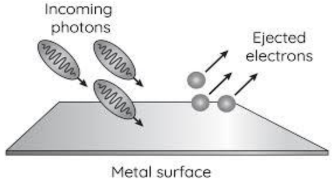
6. *Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt;*
von A. Einstein.

Zwischen den theoretischen Vorstellungen, welche sich die Physiker über die Gase und andere ponderable Körper gebildet haben, und der Maxwellschen Theorie der elektromagnetischen Prozesse im sogenannten leeren Raume besteht ein tiefgreifender formaler Unterschied. Während wir uns nämlich den Zustand eines Körpers durch die Lagen und Geschwindigkeiten einer zwar sehr großen, jedoch endlichen Anzahl von Atomen und Elektronen für vollkommen bestimmt ansehen, bedienen wir uns zur Bestimmung des elektromagnetischen Zustandes eines Raumes kontinuierlicher räumlicher Funktionen, so daß also eine endliche Anzahl von Größen nicht als genügend anzusehen ist zur vollständigen Festlegung des elektromagnetischen Zustandes eines Raumes. Nach der Maxwellschen Theorie ist bei allen rein elektromagnetischen Erscheinungen, also auch beim Licht, die Energie als kontinuierliche Raumfunktion aufzufassen, während die Energie eines ponderabeln Körpers nach der gegenwärtigen Auffassung der Physiker als eine über die Atome und Elektronen erstreckte Summe darzustellen ist. Die Energie eines ponderabeln Körpers kann nicht in beliebig viele, beliebig kleine Teile zerfallen, während sich die Energie eines von einer punktförmigen Lichtquelle ausgesandten Lichtstrahles nach der Maxwellschen Theorie (oder allgemeiner nach jeder Undulationstheorie) des Lichtes auf ein stets wachsendes Volumen ~~sich~~ kontinuierlich verteilt.

On a Heuristic Point of View about the Creation and Conversion of Light

The photoelectric effect



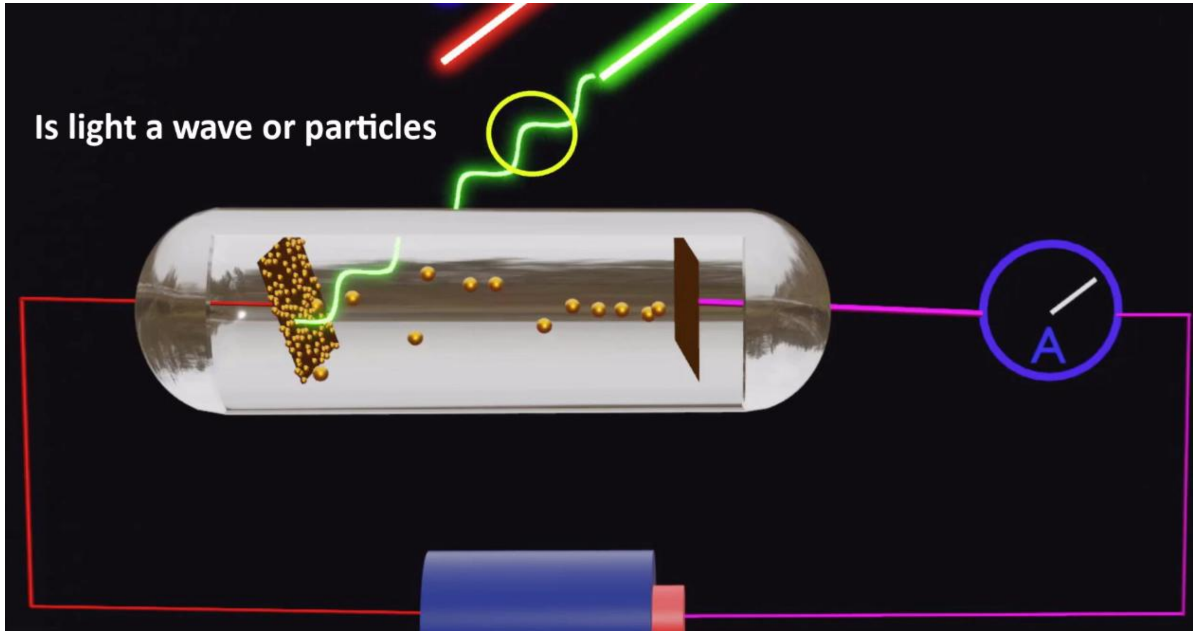


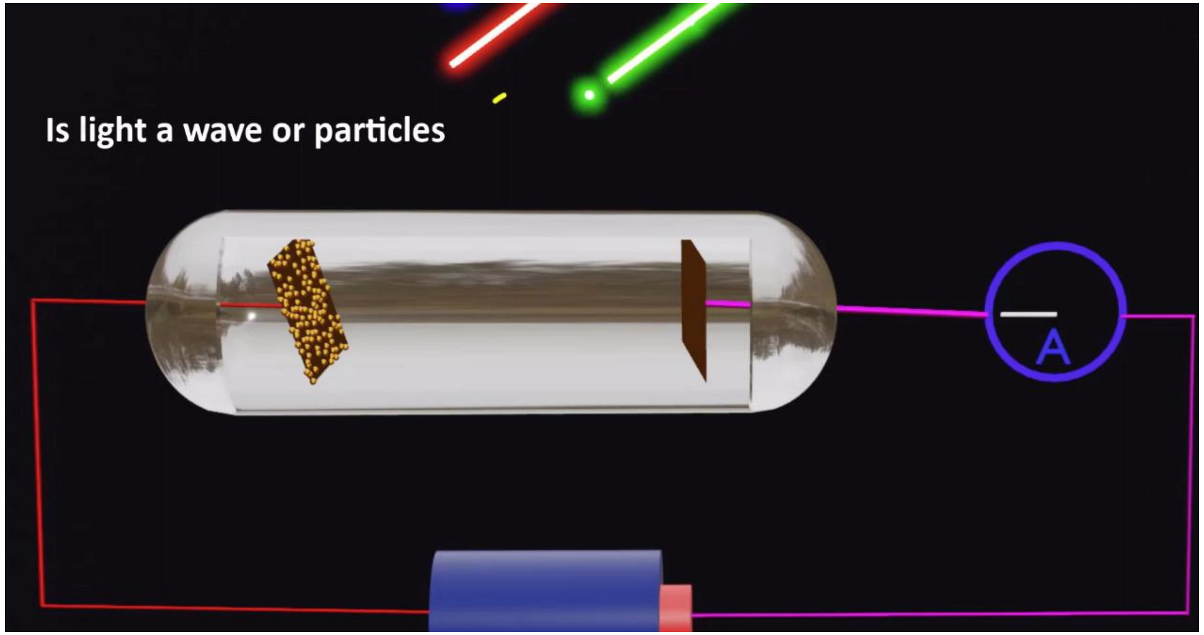
Einstein found:

- Dim high-frequency light knocks out electrons immediately.
- Very bright low-frequency light does nothing.
- Increasing brightness does NOT help if the frequency is too low.

This made no sense for a wave. It was as if light behaved like tiny bullets, not like smooth waves.

A Few Key Takeaways from the Quantum World – The Particle Nature of Light





In 1905, Einstein proposed:
- Light does not deliver energy continuously.
- Light comes in tiny packets (lumps) of energy.
- Each packet carries a fixed amount of energy.

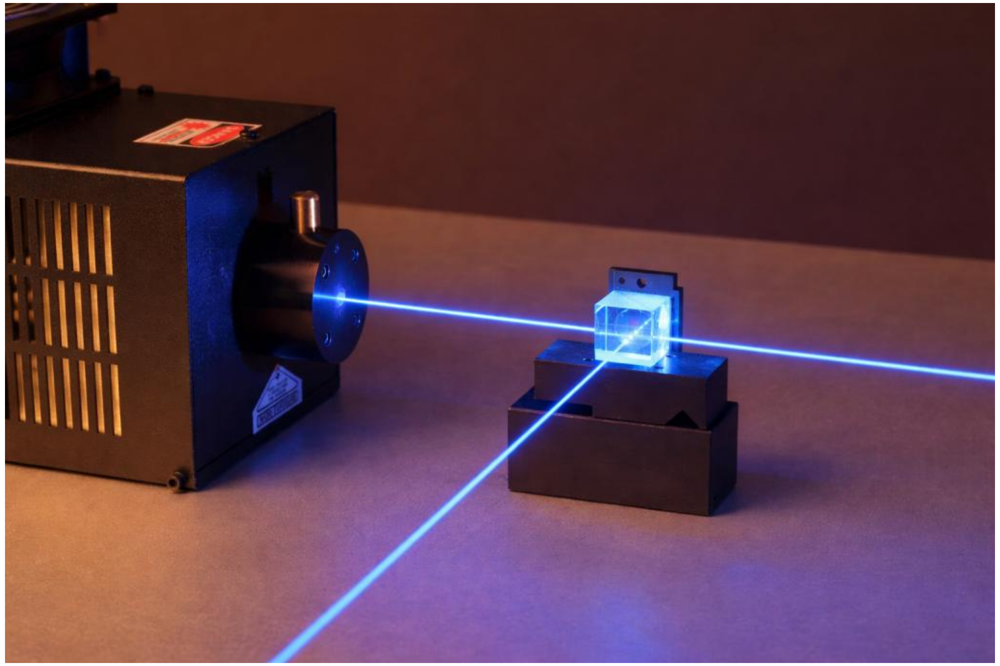
These packets are called photons.

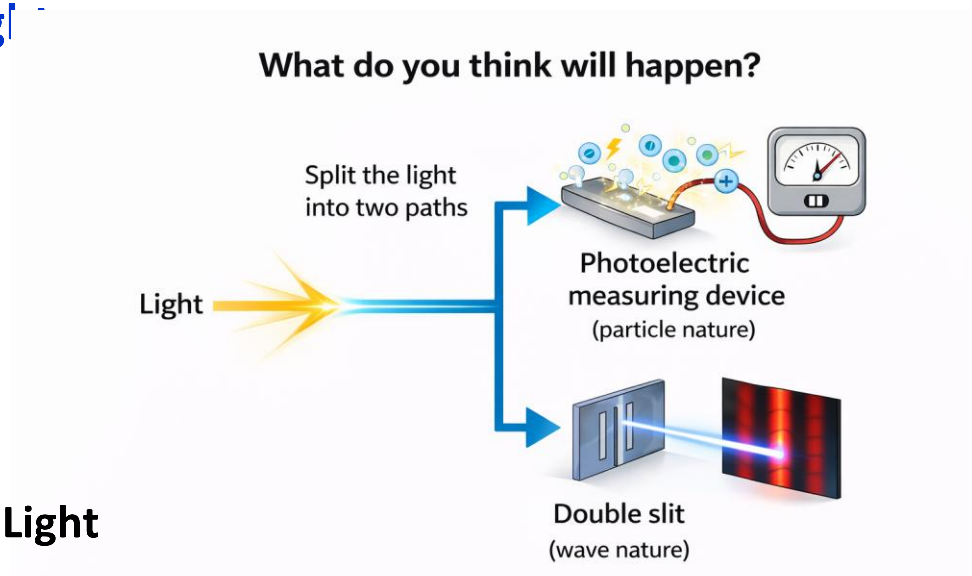
Why This Explained Everything
- Each photon hits one electron.
- If the photon's energy is high enough → the electron escapes.
- If the photon's energy is too low → nothing happens.
- Brighter light means more photons, not more energy per photon.

Suddenly, all observations made sense.

Einstein did not win the Nobel Prize for relativity.
He was awarded the Nobel Prize in Physics for explaining the photoelectric effect using this particle idea of light.

A Few Key Takeaways from the Quantum World – The Nature of Ligl



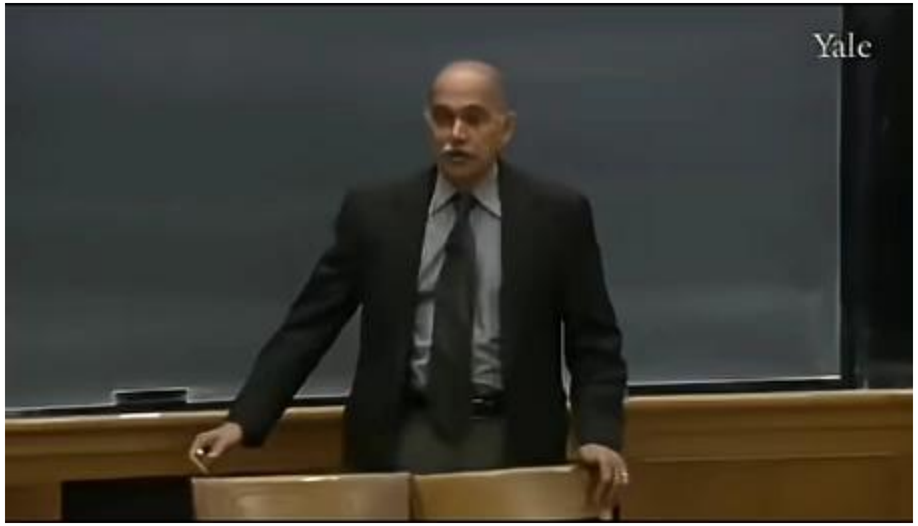


The Dual Nature of Light

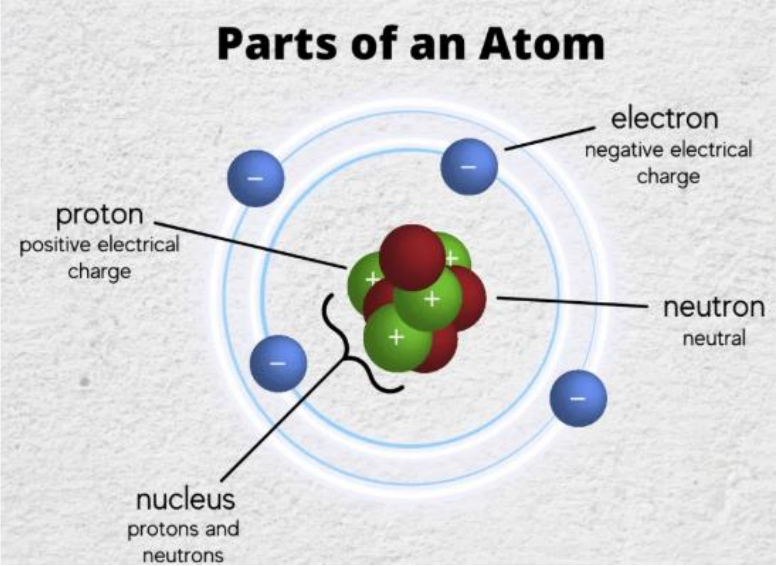
- Light produces interference fringes in a double slit setup → showing clear wave behavior.
- Light produces a photoelectric current in a photoelectric device → showing clear particle behavior.
- The same light, when split into two paths (using an optical beam splitter) can show interference on one path and behave like individual photons on another path.
- It almost seems as if light “knows” that it is being watched (measured), and even how it is being measured — behaving like a wave in one path and like particles in another.
- This deeply surprising behavior shattered classical ideas and opened the door to the quantum world in the post-Einstein era

A Few Key Takeaways from the Quantum World – The Nature of Particles

The Quantum world (and the theory which naturally follows, that is Quantum Mechanics) is weirder than any of us can imagine!!

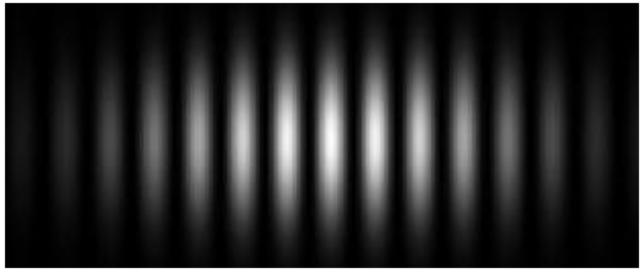


Prof. Ramamurti Shankar
Dept. of Physics, Yale

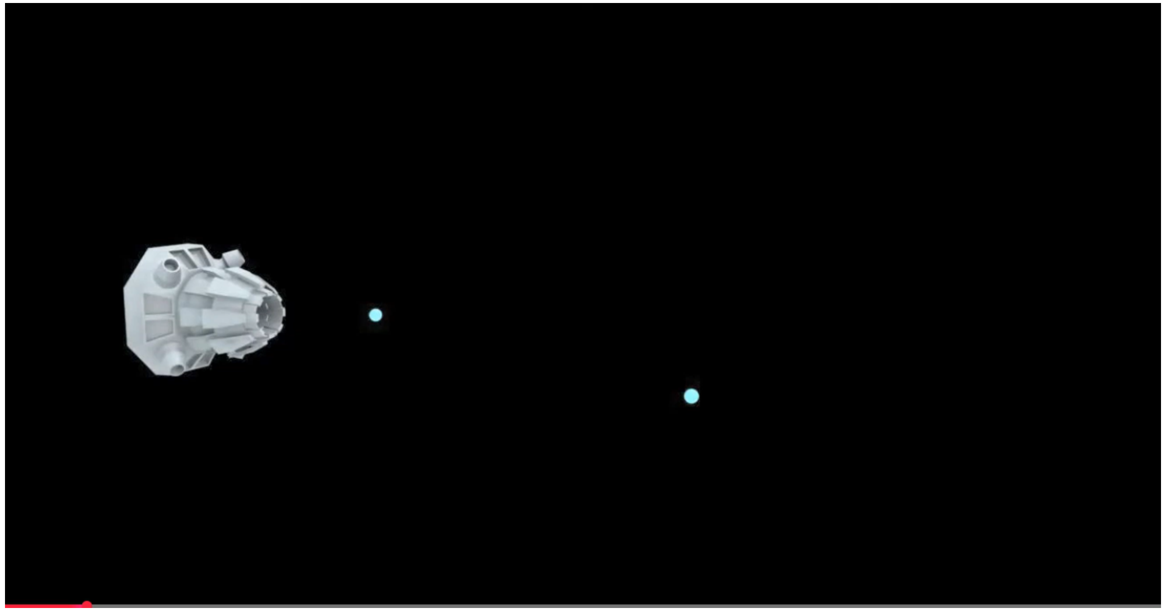


- From school textbooks, we learn that an electron is a fundamental particle of nature. It has mass and is often drawn as a tiny solid sphere with a clear boundary.
- We usually imagine particles (like electrons, protons, neutrons) as small, localized objects, like tiny balls.
- But we just saw that light, which we thought was purely a wave, can also behave like particles (photons).
- This led scientists to ask a bold question: If waves can behave like particles, could particles also behave like waves?
- That question marked the beginning of a completely new way of thinking and pushed physics deeper into the quantum world.

A Few Key Takeaways from the Quantum World – The Nature of Particles

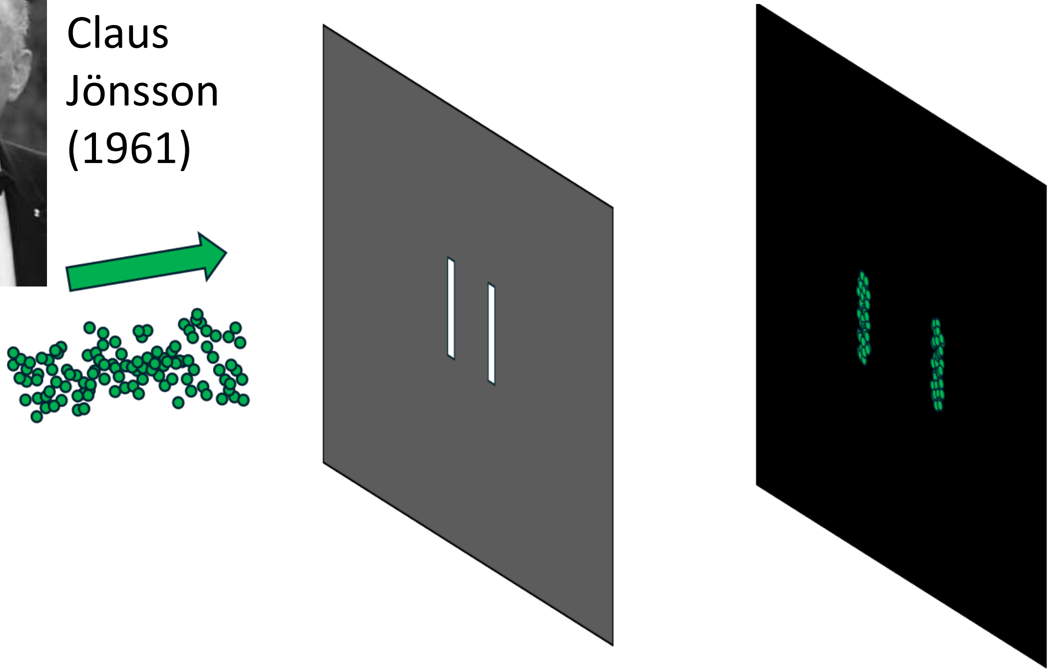


This is how waves behave in a double slit experiment

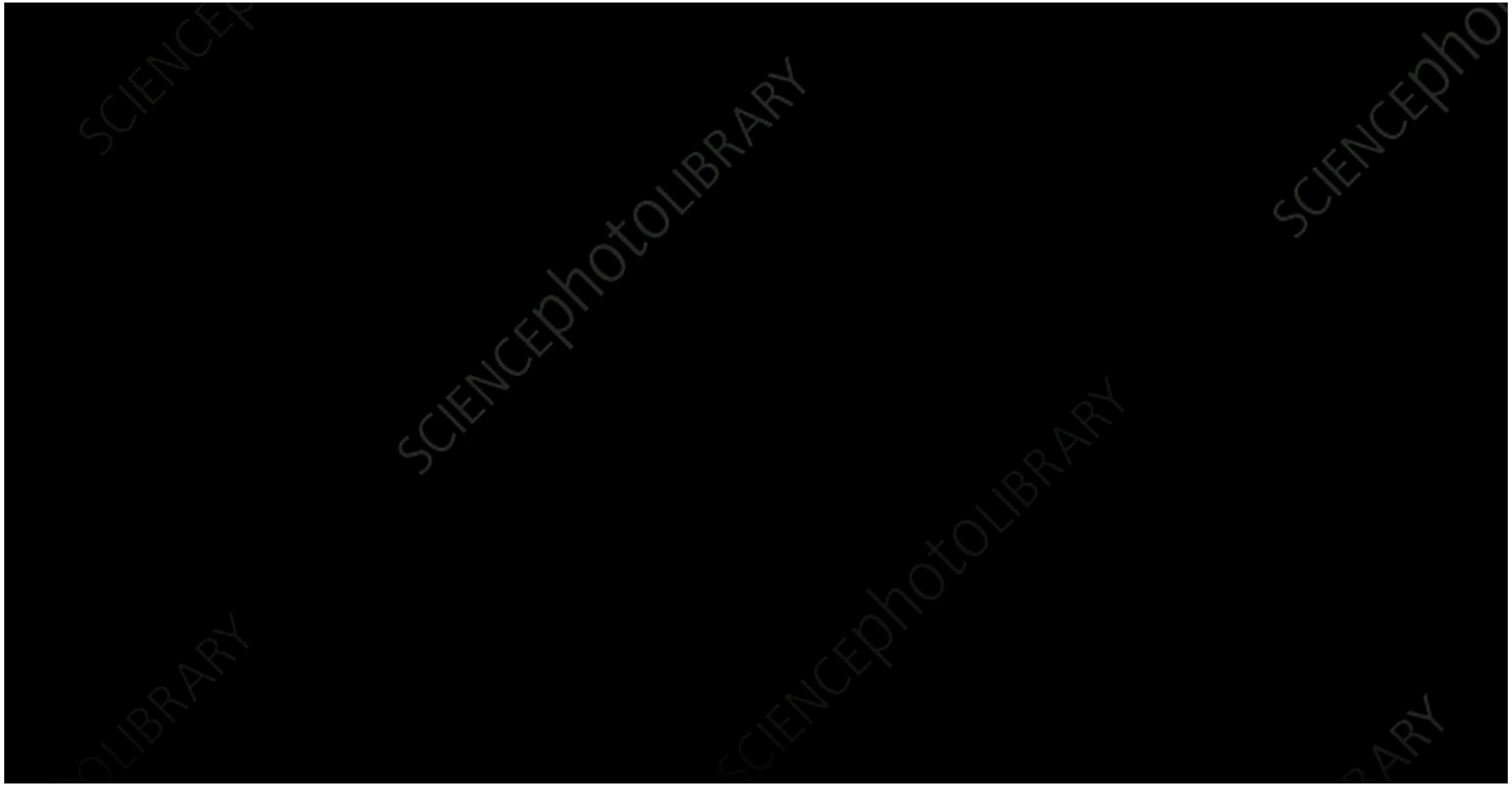


This is how we would expect electrons to behave in a double slit experiment. Since electrons are particles, we would assume that each electron passes through either one slit or the other. As a result, they should pile up directly behind the two slit openings, forming just two bright lines on the screen

Claus Jönsson (1961)

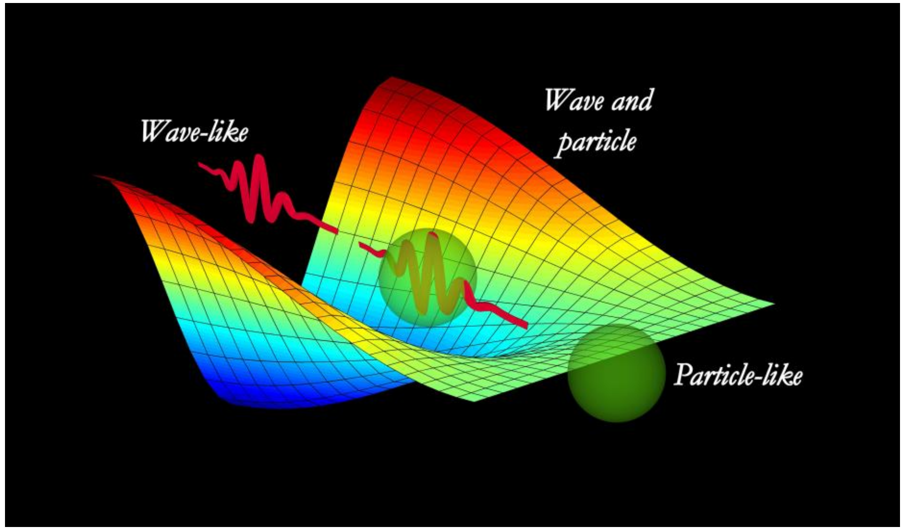


A Few Key Takeaways from the Quantum World – The Nature of Particles



What you are seeing is real experimental data of electrons hitting a fluorescent screen placed behind a double slit apparatus.
Each tiny flash on the screen marks the arrival of a single electron.

A Few Key Takeaways from the Quantum World – The Wave Particle Duality

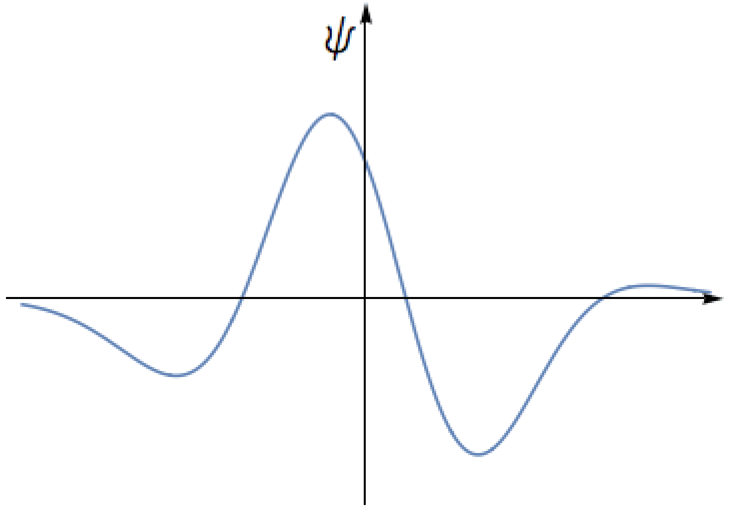


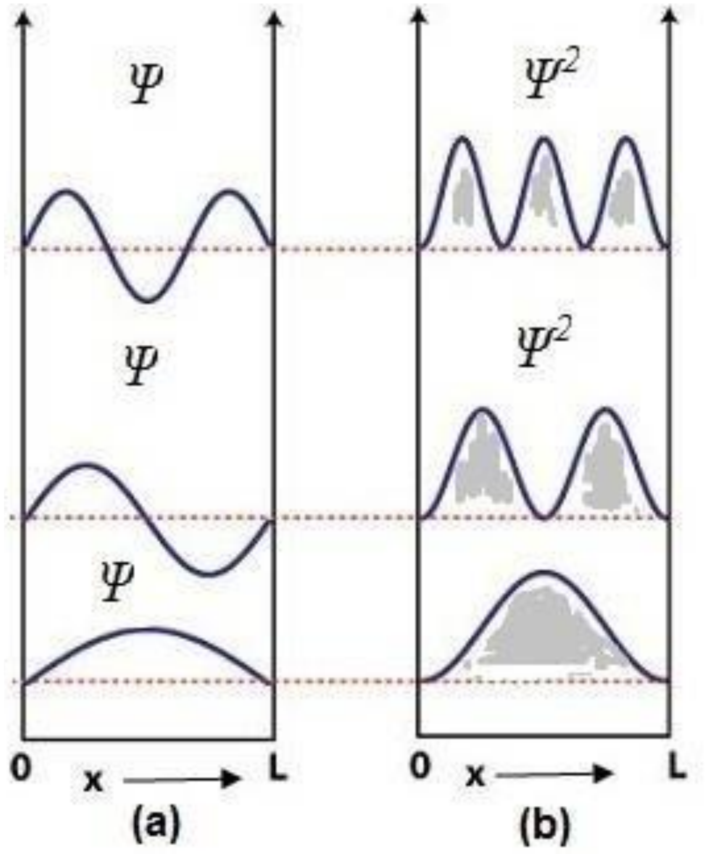
What we call a “particle” is not just a tiny solid object.
At the quantum level, it behaves like a wave spread out in space and if we can carefully control and manipulate this wave inside specially designed circuits (what we call quantum circuits), we can make it perform computations in ways classical particles never could. That control of quantum wave behavior is the very essence of quantum computing.

A Bat has some characteristics that resemble a bird and some other characteristics that resemble a mouse. We don’t call it being a bird and a mouse at the same time. It’s rather a completely different animal which partly resembles a bird and partly a mouse.

A Few Key Takeaways from the Quantum World – The Wave Function of a Particle

In quantum physics, owing to the wave-particle duality, any particular quantum state of particle (like an electron) is described by something called a wave function. We usually write it as $\psi(x)$. Physicist would call it a "Probability wave"



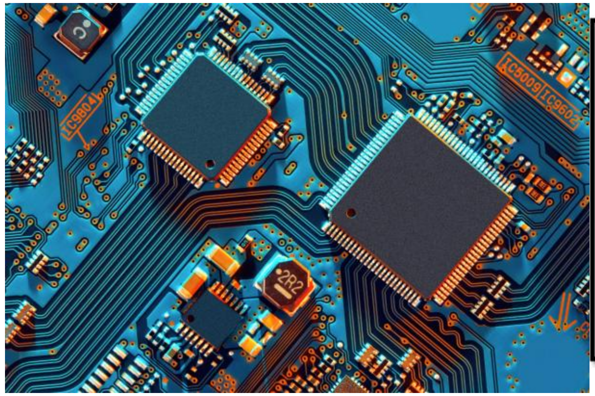


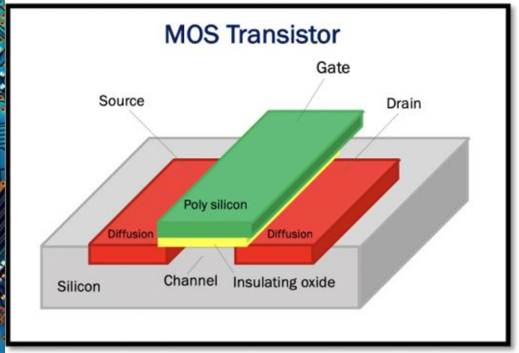
The quantity $|\psi(x)|^2$ gives the probability density of finding the particle at position 'x' in the particular quantum state.

This means the wave function itself is not directly observable but its square modulus tells us the likelihood of detecting the particle at different locations.

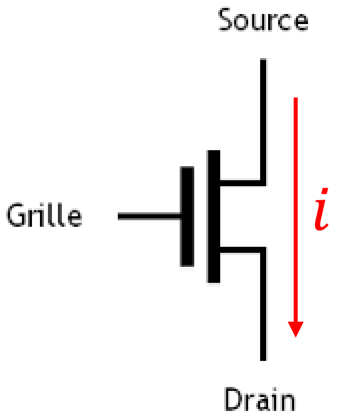
A Few Key Takeaways from the Quantum World – The Probability Wave

What is a Qubit?

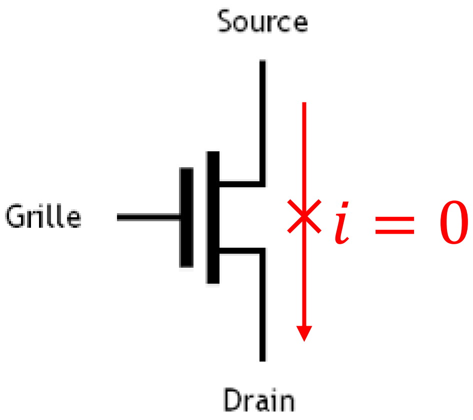




MOSFET transistors are at the heart of modern-day computer chips. They act like a switch, transmitting (classical bit '1') and blocking (classical bit '0') current on application of a gate voltage



Digital -1



Digital - 0

A **qubit** is the quantum version of a classical bit. It is built from a **quantum two-level system**, meaning a system that has two distinct quantum states

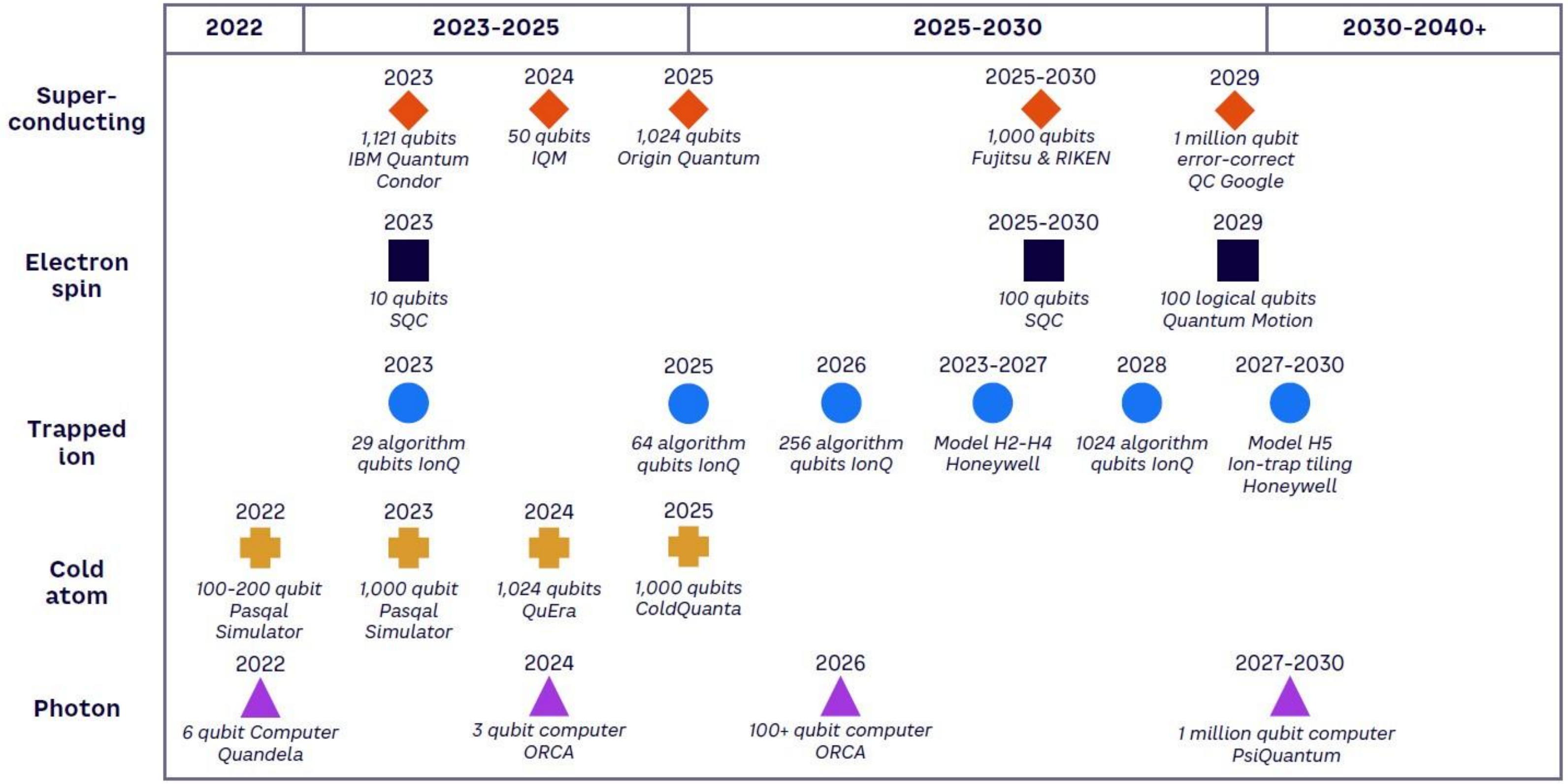
Can a "quantum coin" or rather a hypothetical quantum version of a coin be a qubit?

Can a "quantum die" be a qubit?

Can a "quantum light switch" be a qubit?

Can a "quantum traffic light" be a qubit?

Quantum Computing Roadmap – Qubit Candidates

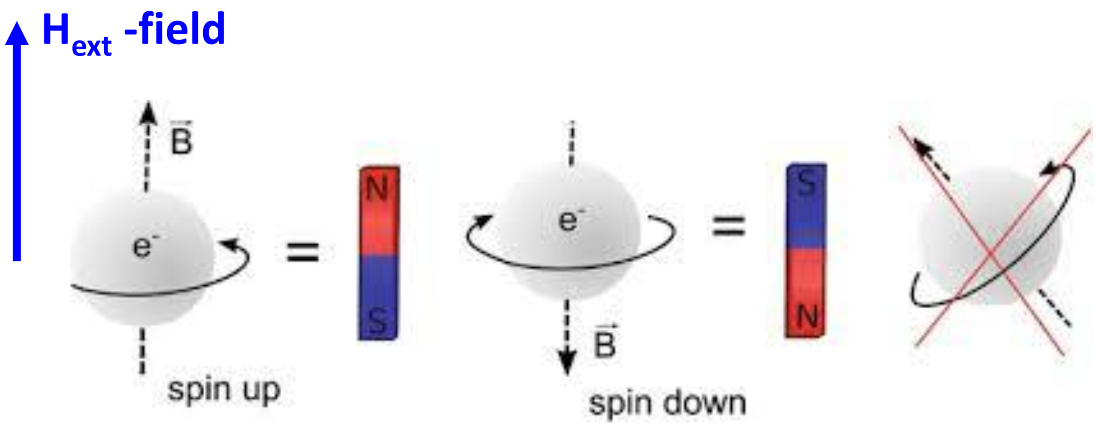


Source: Arthur D. Little, Olivier Ezratty

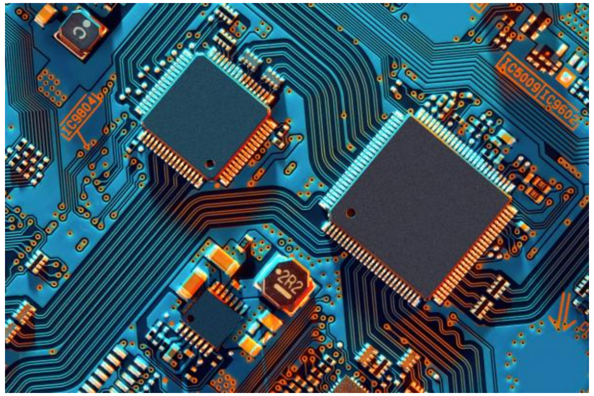
Quantum computing today looks like classical computing in the 1960s — many competing architectures.

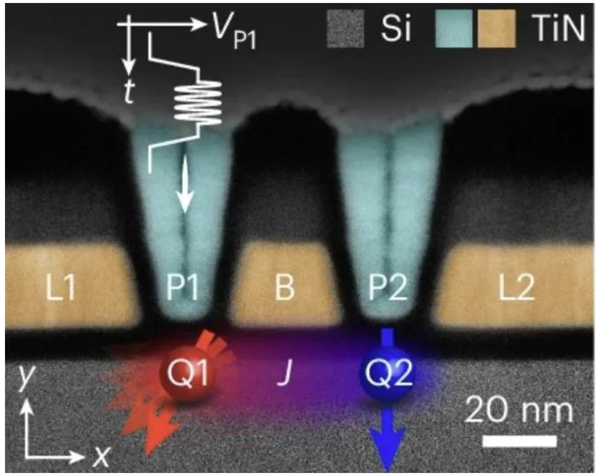
- Multiple quantum hardware platforms are progressing in parallel
- Superconducting systems lead in raw qubit numbers
- Trapped ions and spins emphasize quality and logical qubits
- Photonic and cold-atom platforms aim for long-term scalability
- Timelines reflect engineering difficulty, not optimism

Two-level System: Understanding a Qubit



An electron possesses some intrinsic angular momentum. We call it spin. For a single electron under H-field, the spin can either point up. Or it can point down. **This forms a two-level system**. The spin cannot have any other orientation. We can call the spin-up as the quantum-0 state and the spin-down as the quantum-1 state.

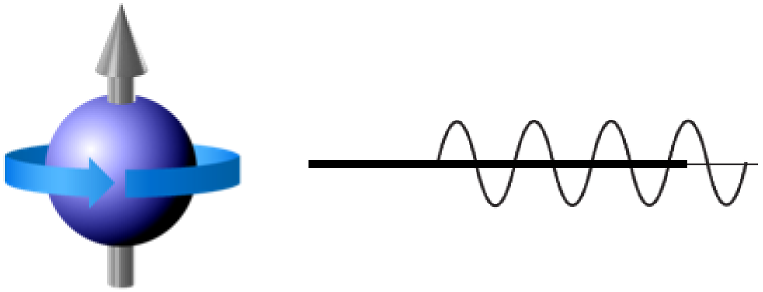




- The same electron that flows through a MOS transistor behaves like a classical particle in everyday circuits. Its quantum wave nature is hidden by noise, temperature, and interactions.
- To observe its quantum (wave-like) behavior, we must isolate and control it using specially designed quantum circuits and carefully engineered experimental setups.
- These systems operate under extreme conditions (very low temperatures, precise electromagnetic control) to prevent the fragile quantum state from being disturbed.
- Only under such controlled conditions does the electron reveal its wave nature and learning to manipulate that wave is what allows us to perform quantum computation.

Two-level System: Understanding a Qubit

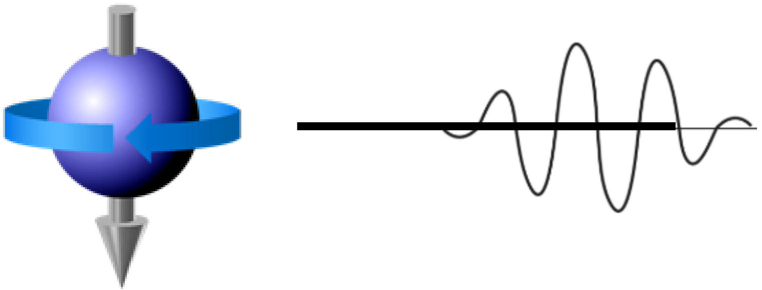
spin up



$|\mathbf{0}\rangle$

spin down

$$|\boldsymbol{\psi}\rangle = \boldsymbol{\alpha}|\mathbf{0}\rangle + \boldsymbol{\beta}|\mathbf{1}\rangle, \textbf{ where } \{\boldsymbol{\alpha}, \boldsymbol{\beta}\} \in \mathbb{C}$$



$|\mathbf{1}\rangle$

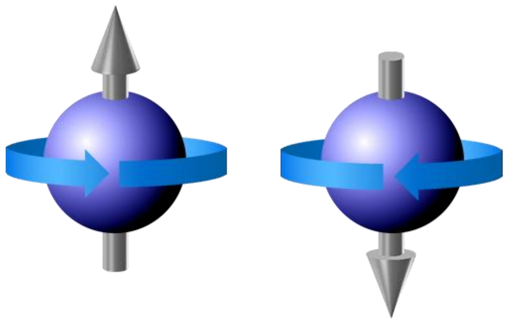
- An electron behaves like a tiny magnet in an external magnetic field, and it can point in only two directions: aligned (up) or anti-aligned (down) — giving us a natural two-level system ($|0\rangle$ and $|1\rangle$).
- In the quantum world, each of these states is described by a wave.
- Since waves can overlap, we can create a superposition state
- $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ which is a linear combination of the two quantum states, $|0\rangle$ and $|1\rangle$ (later we will call these $|0\rangle$ and $|1\rangle$ as basis states).
- This is possible only because quantum states behave like waves, something impossible in classical systems.
- In a classical MOS transistor, a state cannot be written as "α × current flowing + β × current not flowing". NO. It must be either 0 or 1, never both together.

Classical bits are like either HEAD or TAIL

Qubits can be prepared in a state that is both HEAD and TAIL at the same time

Motivation: The Magic is in Superposition

spin up **spin down**



$|0\rangle$ $|1\rangle$

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $\{\alpha, \beta\} \in \mathbb{C}$

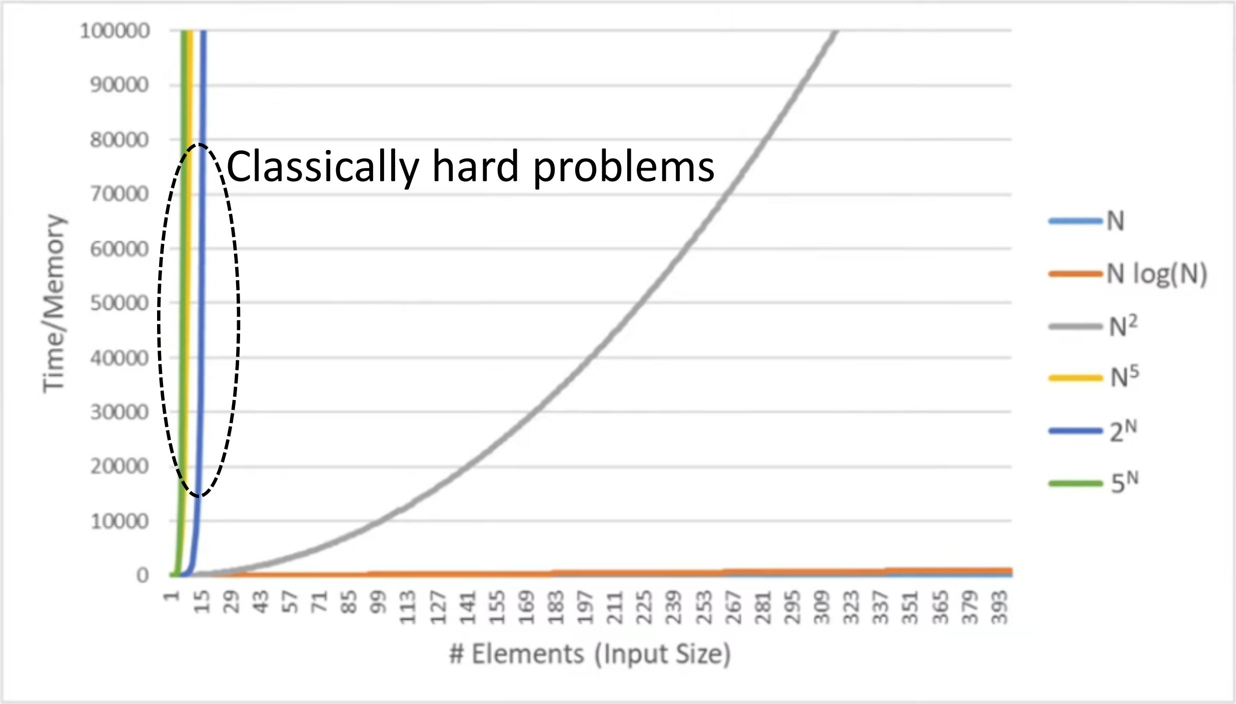
1 qubit → 2 numbers → 2 bits of information

1 classical bit → 1 bit of information

$|\psi\rangle = \alpha|0\rangle|0\rangle + \beta|0\rangle|1\rangle + \gamma|1\rangle|0\rangle + \delta|1\rangle|1\rangle$, $\{\alpha, \beta, \gamma, \delta\} \in \mathbb{C}$

2 qubits prepared in a state of superposition
→ 4 numbers → 4 bits of information

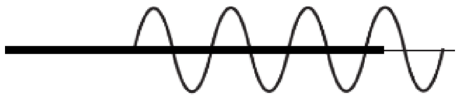
n qubits prepared in a state of superposition
→ 2^n bits of information



Quantum Computers are not a replacement to classical computers. They are useful to solve only those problems that are beyond the scope of the most advanced supercomputers today.

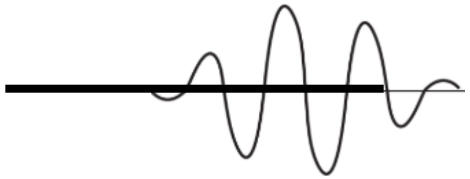
Quantum computers are simple not a better or more powerful version of classical computers (they are a radical new technology. Just like a light bulb is not a better version of a candle, it's a whole new and different technology altogether

Superposition of States



$|\mathbf{0}\rangle$

$$|\boldsymbol{\psi}\rangle = \boldsymbol{\alpha}|\mathbf{0}\rangle + \boldsymbol{\beta}|\mathbf{1}\rangle,$$
where $\{\boldsymbol{\alpha}, \boldsymbol{\beta}\} \in \mathbb{C}$



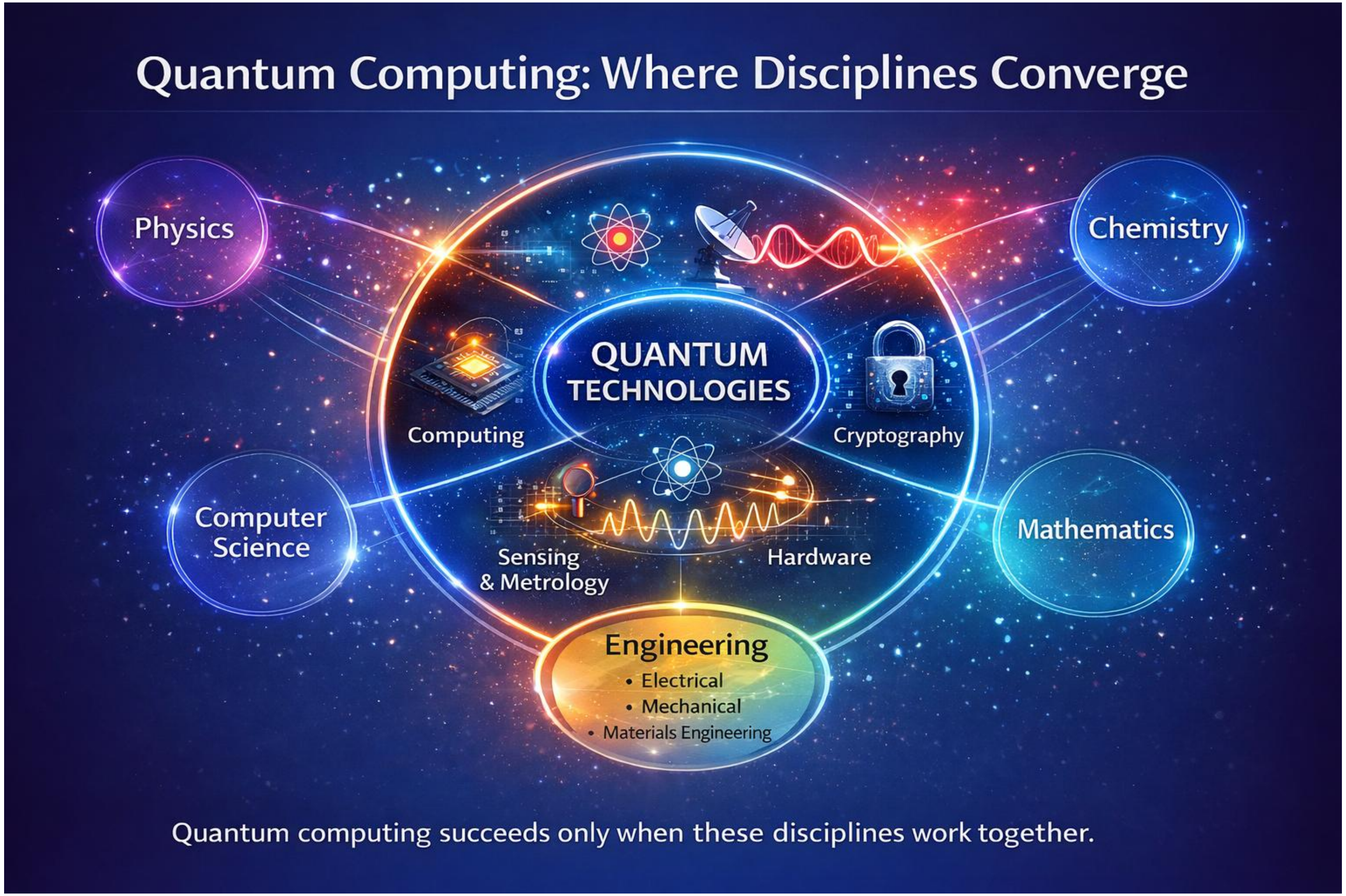
$|\mathbf{1}\rangle$

- Think of a spinning coin: while it is spinning, it is not simply heads or tails. It is in a kind of "both at once" state. A qubit in superposition is similar: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
- In quantum experiments, we can prepare a qubit in such a superposition and even manipulate it carefully inside quantum circuits.
- But the moment we measure the qubit, the superposition collapses. Just like the spinning coin falling to either heads or tails.
- The probability that it collapses to $|0\rangle$ is $|\alpha|^2$, and the probability that it collapses to $|1\rangle$ is $|\beta|^2$.
- We cannot simply "look" at a qubit to measure it. Measurement requires a physical interaction - shining light on it, driving a current, or coupling it to an external device, and this interaction disturbs the system, causing the superposition to collapse into one of the basis states
- Measurement converts quantum information into a definite classical outcome and this controlled collapse is a central feature of quantum computing

Example: I create the following superposition of states: $|\psi\rangle = \sqrt{0.3}|0\rangle + \sqrt{0.7}|1\rangle$. If I try to perform a measurement of this state, I will find my qubit to be in state $|0\rangle$ with a probability 0.3 and in the state $|1\rangle$ with a probability 0.7. Never will I ever measure it to be in the superposition state of $\sqrt{0.3}|0\rangle + \sqrt{0.7}|1\rangle$

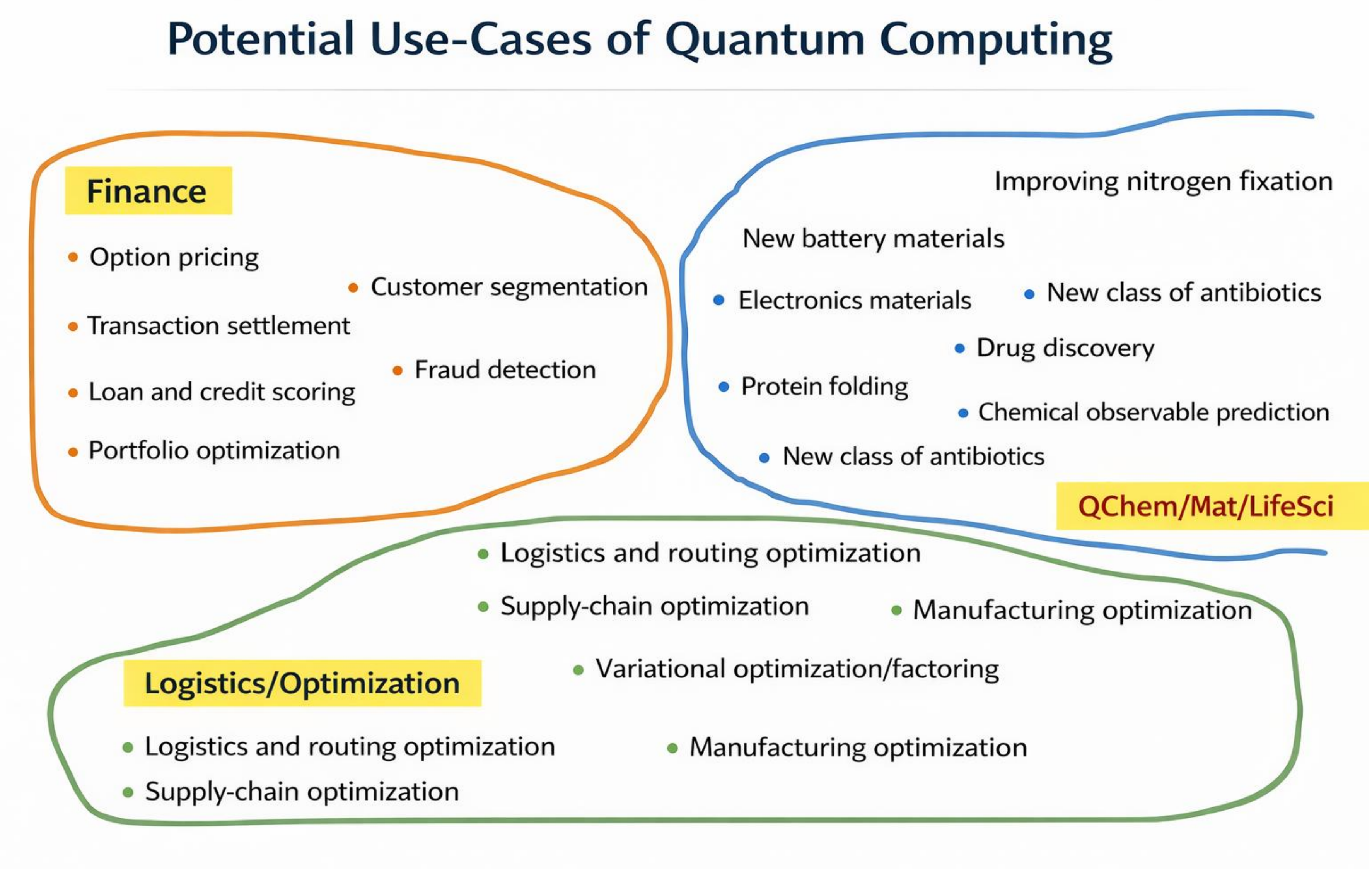
Also, note that the total probability has to be 1. This means, for any given superposition $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $|\alpha|^2 + |\beta|^2 = 1$ has to be satisfied. For example, $|\psi\rangle = \sqrt{0.5}|0\rangle + \sqrt{0.7}|1\rangle$ is not a valid quantum state, since the total probability is exceeding 1.

Application Space



Quantum computing is not a field you enter. Rather, it's a place where many fields collide

Industry Used Cases



Summary

- Quantum entities such as electrons, photons, and neutral atoms, when properly engineered and controlled, exhibit behavior that is partly wave-like and partly particle-like. This dual nature is not a curiosity. It is the foundation of quantum technologies.

- The wave nature of quantum entities allows them to exist in a state of superposition, meaning a quantum state can be expressed as a linear combination of basis states. In the context of quantum computing, this is what allows a qubit to represent more than just a single classical value at a time.

- The power of quantum computing comes directly from superposition. If a system cannot create superposition or if the superposition is lost due to noise or premature measurement, then there is no quantum advantage. A quantum computer without superposition behaves like a classical machine.

- Superposition, however, cannot be directly measured. The moment a qubit in a superposed state is measured, it collapses into one of the basis states, according to a probability rule.

- **A key principle of quantum algorithms is that no measurement should be performed during the computation. Measurement is deferred to the very end, and the algorithm must be designed so that the final state yields meaningful information when measured. If the system remains in an unstructured superposition at the end, measurement will destroy the information.**